

Julio Quiroga Galan

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-Spanish (Native) -English (Bilingual) -French (Basic)

Education

Michigan State University

East Lansing, MI

-Master of Science in Electrical & Computer Engineering

4.0 GPA/Expected Dec 2026

University of California, Santa Cruz

Santa Cruz, CA

-Bachelor of Science in Robotics Engineering & Electrical Engineering Minor (CUM LAUDE)

June 2025

Work History

Graduate Researcher, Cyber-Physical Systems Lab

East Lansing, MI

Sep, 2025 - Present

- Researching LLM-guided Multi-Agent Reinforcement Learning to improve collaboration and control efficiency to achieve high-level goals.
- Designing and analyzing optimal & nonlinear control policies (LQR/LQG, MPC, Lyapunov-based) to ensure stability, safety, and performance.
- Building Python Deep Neural Networks that turn LLM outputs into verifiable controllers, with on-robot execution and closed-loop telemetry.

TA for Science & Engineering Entrepreneurship

East Lansing, MI

Sep 2026 - Present

- Selected as a teaching assistant after strong performance in the course the previous year.
- Supported students in turning engineering innovations into viable business opportunities through market analysis and pitch development.
- Guided students through the mathematics in finance and business, including TVM, income statements, investments, and cost structure.

Founder, Real Robotics

East Lansing, MI

Dec, 2025 - Present

- Designed and simulated autonomous drone systems for search-and-rescue missions in support of first responders
- Performed market and feasibility analyses to evaluate business viability and develop investor-facing reports and pitch materials
- Won elevator pitch competitions and secured funding to advance startup development

R&D Engineer, Ikergune

Donostia-San Sebastian, Spain

Jun, 2025 - Aug, 2025

- Integrated industrial PLCs with IO-Link sensors and actuators over PROFINET for real-time control and data acquisition.
- Built and customized Linux OS images with the Yocto Project (kernel features, device trees, layers) for robotics hardware optimization.
- Programmed and debugged embedded System communication protocols (SPI, UART, I2C, CAN) for automated systems.

Sensor Development & Integration Engineer, SYLVAE

Santa Cruz, CA

Sep, 2024 - Jun, 2025

- Led LiDAR and temperature sensing for a ground wildfire-risk mitigation platform on ESP32/Linux.
- Designed and tested PCBs (Cadence Allegro), state machines for feedback response, and C/C++ data acquisition.
- Analyzed point clouds in CloudCompare and produced calibrated field measurements.
- Coordinated with other teams to integrate the sensors into the final product.

Undergraduate Researcher, Hare Lab

Santa Cruz, CA

Jun, 2024 - Sep, 2024

- Prototyped a 4+1 VTOL drone for humanitarian missions.
- Ran simulations in ArduPilot and developed multi-sensor perception for resource and survivor detection.

Skills

Python, MATLAB, C / C++, DNN, SolidWorks, Linux, PCB design, Cadence Allegro, KiCad, Reinforcement learning, Control systems, Nonlinear control, Optimal control, LQR, MPC, Electromechanical systems, PLC integration, SPI, UART, I2C, CAN, LiDAR, CloudCompare

Projects ([Julio's Engineering Portfolio](#))

Inverted Pendulum (Applied Feedback Control):

- Derived Lagrangian and state-space model, verified controllability/observability, and designed LQR controller and estimator.
- Executed MATLAB inverted-pendulum simulations ([Inv Pend Sim](#)) to evaluate motor and hardware limit analysis.
- Translated the design to embedded C on an MCU using an electromechanical plant model for real-time control.

Mechatronics Robot Design:

- Led a 10-week electro-mechanical system that required 3D design SolidWorks, MCU embedded C state machines, A/D & D/A, and filters.
- Sensor development with photo-transistor, band-pass filters for tower detection, tank circuit for door localization, and black-tape navigation.
- Software design of hierarchical state machines and embedded communication protocols for hardware interaction.
- Record of 37/40 balls collected and 33/40 deposited under 2 minutes at a designated door.

Formula SAE (Electric Race Car)

- Developed the tractive system for an electric race car
- Designed KiCad PCBs for pre-charge/discharge with MOSFET switching, measurement-point control, and Manual Service Disconnect
- Worked with a high-voltage system of 120 V / 700 A accumulator, implementing protection and control features.

Honors

Burgess Institute of Entrepreneurship Launch Program	East Lansing, MI	March 2026
-Recognized and entered into the exclusive Launch program of BIE to continue developing a startup recognized in competitions and raised funds.		

Engineering Graduate Leadership Fellowship	East Lansing, MI	Fall 2025 - Present
-Recognized for exceptional academic and leadership achievements, and for delivering engineering outreach talks		

Engineering Honor Society (Tau Beta Pi)	Santa Cruz, CA & East Lansing, MI	Spring 2023 - Present
-Recognized for exceptional academic achievement and leadership in engineering, ranking in the top 5% of my class		